

Camera Model

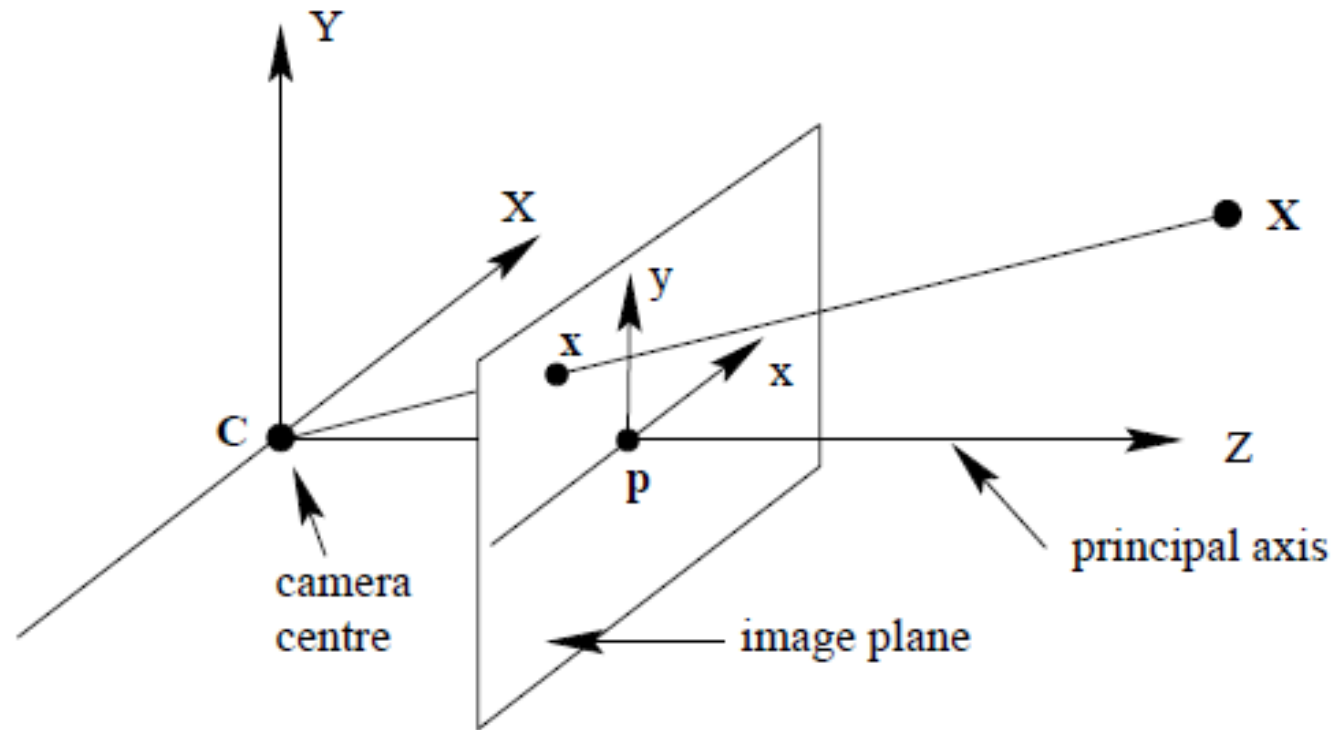
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Outline

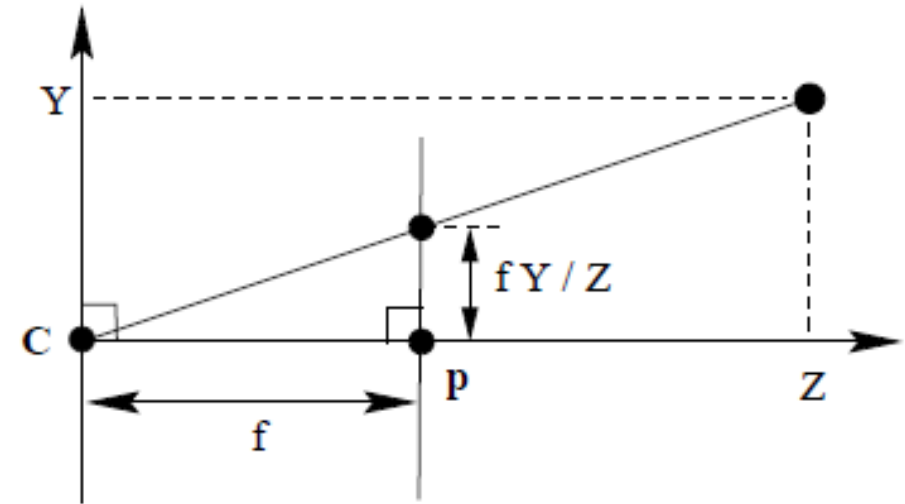
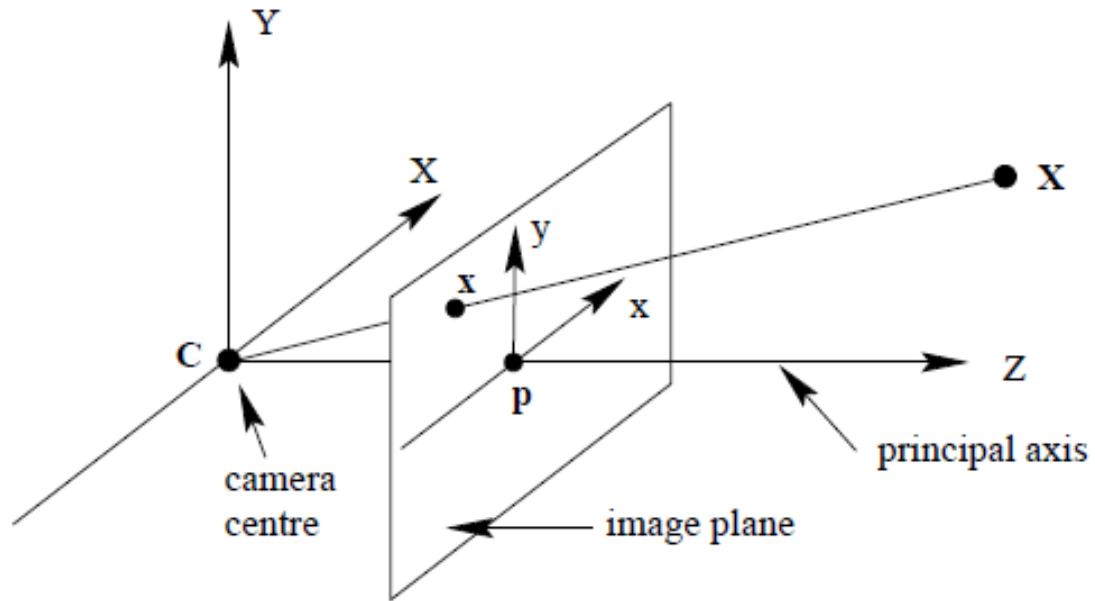
- Pinhole Camera Model
- Lens Distortion

Pinhole Camera Geometry



- C is the camera center, and p is the principal point.
- The camera center C is placed at the world-coordinate origin.
- The principal point p is placed at the image-coordinate origin.

World to Image Coordinates



$$\left(\frac{fX}{Z}, \frac{fY}{Z}\right) \Rightarrow \underbrace{\begin{bmatrix} fX \\ fY \\ Z \end{bmatrix}}_{\tilde{\mathbf{X}}} = \underbrace{\begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{K}} \underbrace{\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}}_{\tilde{\mathbf{X}}_{cam}}$$

Homogeneous Coordinates

- Add a third/fourth dimension to each 2D/3D point:

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homogeneous image
coordinates

$$(X, Y, Z) \Rightarrow \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

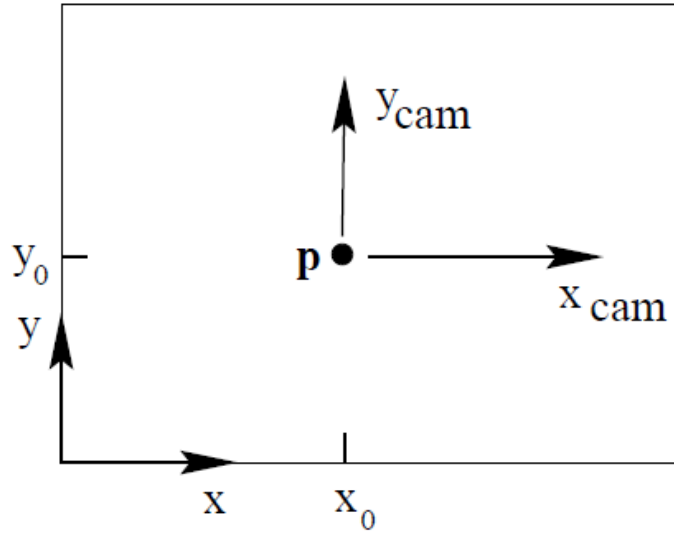
Homogeneous world
coordinates

- To get real 2D/3D coordinates, divide by w:

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow \left(\frac{x}{w}, \frac{y}{w} \right)$$

$$\begin{bmatrix} X \\ Y \\ Z \\ w \end{bmatrix} \Rightarrow \left(\frac{X}{w}, \frac{Y}{w}, \frac{Z}{w} \right)$$

Principal Point Offset



- Previously, we assume the origin of coordinates in the image plane is at the principal point $\mathbf{p}$.
- In practice, it may not be.

$$\underbrace{\begin{bmatrix} fX + Zx_0 \\ fY + Zy_0 \\ Z \end{bmatrix}}_{\tilde{\mathbf{X}}} = \underbrace{\begin{bmatrix} f & 0 & x_0 \\ 0 & f & y_0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{K}} \underbrace{\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}}_{\tilde{\mathbf{X}}_{cam}} \Rightarrow \tilde{\mathbf{x}} = \mathbf{K}[\mathbf{I} \quad \mathbf{0}]\tilde{\mathbf{X}}_{cam}$$

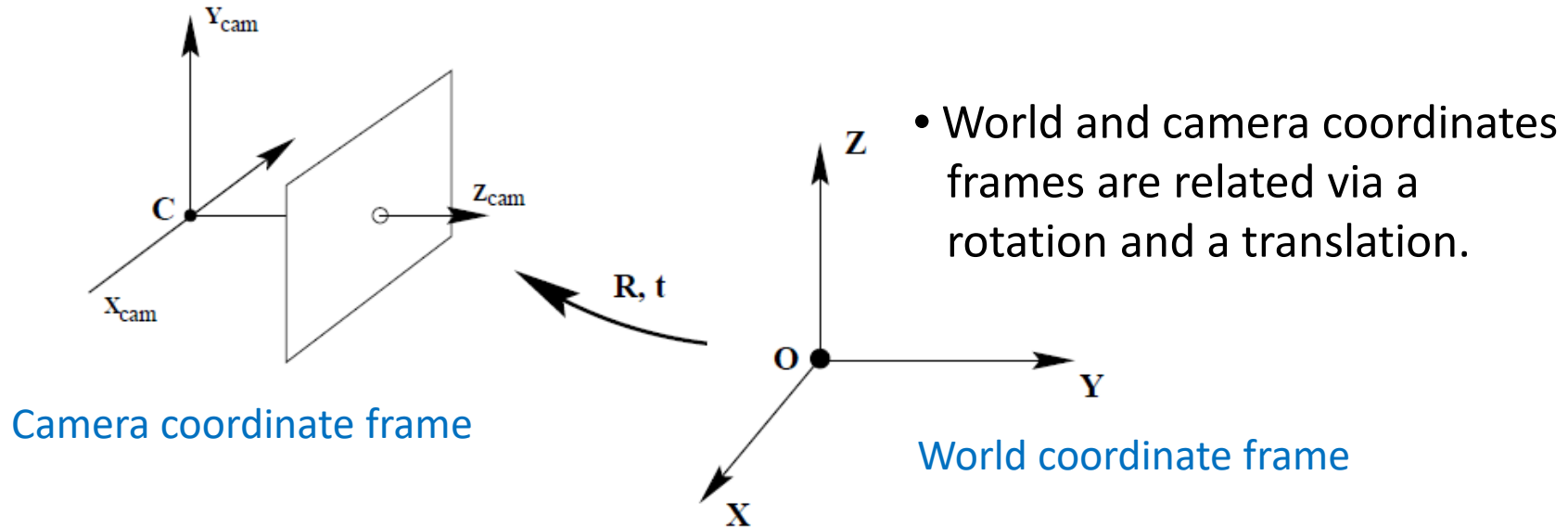
CCD Cameras and Skew

- For CCD cameras, the scale factors in each axial direction may not be equal.
- Thus, the diagonal entries of **K** can be of different values: α_x and α_y .
- The skew parameter s will be nonzero if x - and y -axes are not perpendicular, which is the case when a photograph is re-photographed.

$$\mathbf{K} = \begin{bmatrix} \alpha_x & s & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Notice that **K** is upper triangular.

Camera Rotation and Translation



$$\mathbf{X}_{cam} = \mathbf{R}(\mathbf{X} - \mathbf{c}) \Rightarrow \tilde{\mathbf{X}}_{cam} = \begin{bmatrix} \mathbf{X}_{cam} \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R} & -\mathbf{R}\mathbf{c} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{X} \\ 1 \end{bmatrix}$$

$$\tilde{\mathbf{x}} = \mathbf{K}[\mathbf{I} \ 0] \tilde{\mathbf{X}}_{cam} = \mathbf{K}[\mathbf{I} \ 0] \begin{bmatrix} \mathbf{R} & -\mathbf{R}\mathbf{c} \\ 0 & 1 \end{bmatrix} \tilde{\mathbf{X}}$$

$$\tilde{\mathbf{x}} = \underbrace{\mathbf{K}[\mathbf{R} \ \mathbf{t}]}_{\mathbf{P}} \tilde{\mathbf{X}}, \text{ where } \mathbf{t} = -\mathbf{R}\mathbf{c}$$

Pinhole Camera Model

Intrinsic Matrix Extrinsic Matrix

$$\lambda \tilde{\mathbf{x}} = \overbrace{\mathbf{K}}^{\text{Intrinsic Matrix}} \overbrace{[\mathbf{R} \quad \mathbf{t}]}^{\text{Extrinsic Matrix}} \tilde{\mathbf{X}}$$

$\tilde{\mathbf{x}}$: Homogeneous image coordinates: $(x, y, 1) \in \mathbb{R}^3$

$\tilde{\mathbf{X}}$: Homogeneous world coordinates: $(X, Y, Z, 1) \in \mathbb{R}^4$

$\mathbf{K}$: Intrinsic matrix $\in \mathbb{R}^{3 \times 3}$

$\mathbf{R}$: Rotation matrix $\in \mathbb{R}^{3 \times 3}$

$\mathbf{t}$: Translation vector $\in \mathbb{R}^3$

λ : Scaling factor $\in \mathbb{R}$

Research Topics

- Given $\mathbf{K}$, $(\mathbf{R}, \mathbf{t})$, $\tilde{\mathbf{X}}$, compute $\tilde{\mathbf{x}}$.
 - 3D-to-2D Projection
- Given $\mathbf{K}$, $\tilde{\mathbf{X}}$, $\tilde{\mathbf{x}}$, compute $(\mathbf{R}, \mathbf{t})$.
 - Perspective-n-Point Problem
- Given $\tilde{\mathbf{X}}$, $\tilde{\mathbf{x}}$, compute $\mathbf{K}$, $(\mathbf{R}, \mathbf{t})$.
 - Camera Calibration
- Given $\mathbf{K}$, $\tilde{\mathbf{x}}$, compute $\tilde{\mathbf{X}}$, $(\mathbf{R}, \mathbf{t})$.
 - 3D Reconstruction

Camera Model

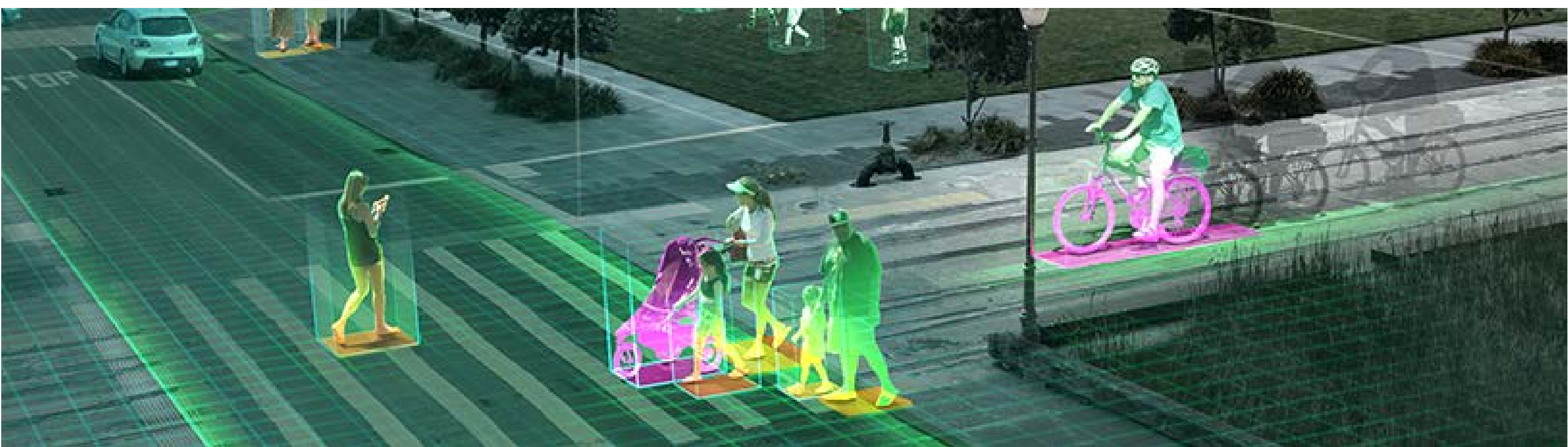
$$\lambda \tilde{\mathbf{x}} = \mathbf{K}[\mathbf{R} \ \mathbf{t}] \tilde{\mathbf{X}}$$

Computer Vision

$$\lambda \tilde{\mathbf{x}} = \mathbf{K}[\mathbf{R} \quad \mathbf{t}] \tilde{\mathbf{X}}$$



Vision



Computer Graphics

$$\lambda \tilde{\mathbf{x}} = \mathbf{K}[\mathbf{R} \quad \mathbf{t}] \tilde{\mathbf{X}}$$

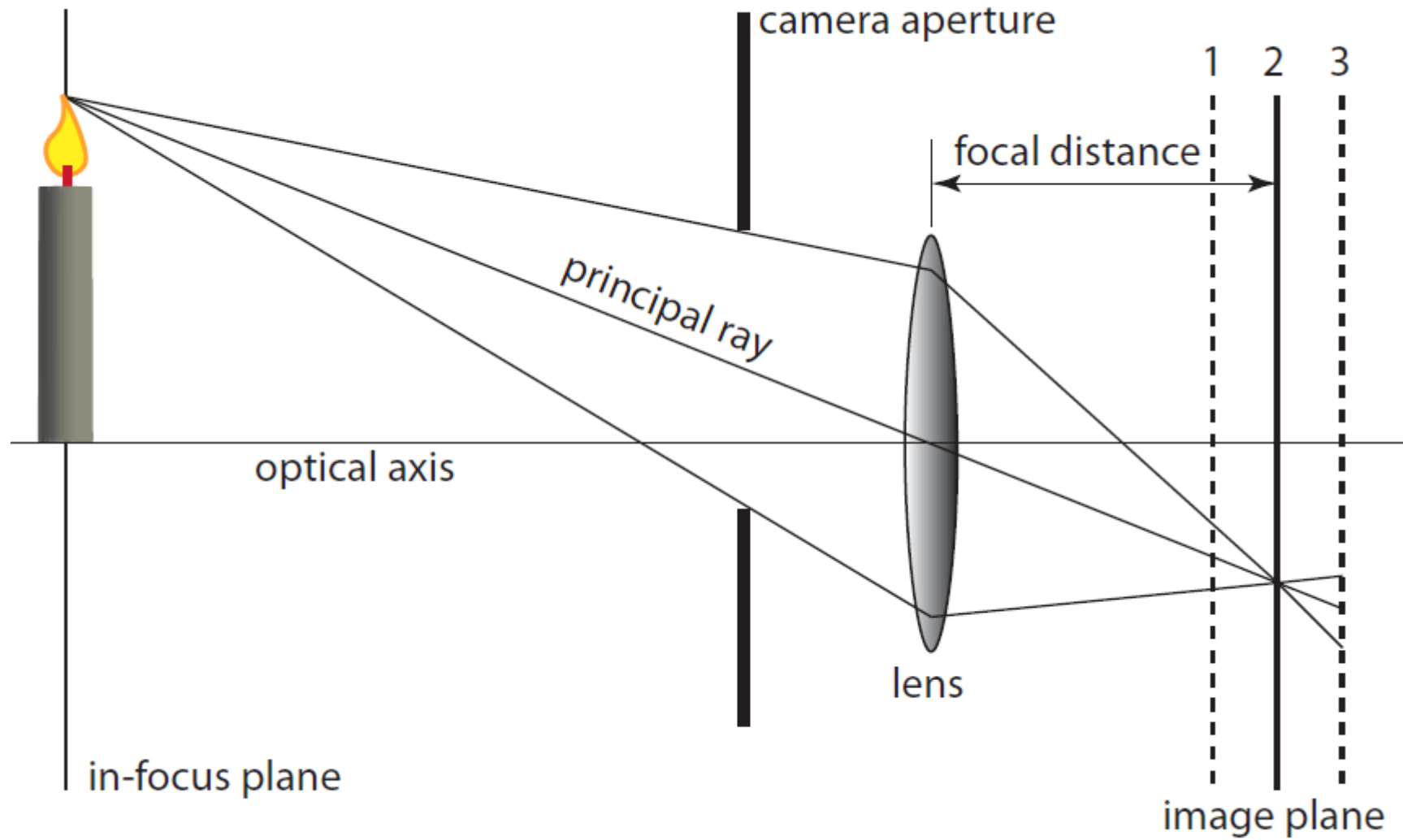
←
Graphics



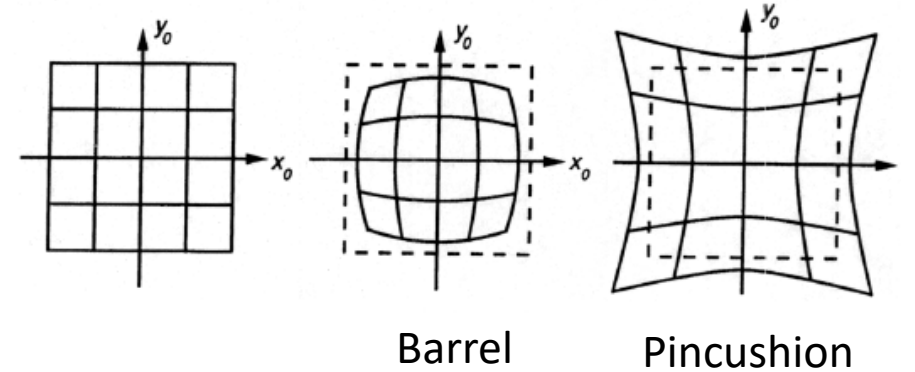
Image Credit:
Trevorgoodchild



Lenses



Lens Distortion



- Radial distortion:

$$x_{distorted} = x(1 + k_1r^2 + k_2r^4 + k_3r^6)$$

$$y_{distorted} = y(1 + k_1r^2 + k_2r^4 + k_3r^6)$$

- Tangential Distortion:

$$x_{distorted} = x + (2p_1xy + p_2(r^2 + 2x^2))$$

$$y_{distorted} = y + (p_1(r^2 + 2y^2) + 2p_2xy)$$

- Distortion coefficients: k_1, k_2, p_1, p_2, k_3
 - Perform camera calibration to compute distortion coefficients